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### Semiconductor device

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#### Abstract

A semiconductor device includes a base, a matching circuit including a substrate, a ground layer, and a signal line, wherein a width of the signal line on a first end side of the substrate is smaller than a width of the substrate and larger than that of the signal line on a second end side, and a distance between the ground layer and the signal line on the first end side is larger than a distance therebetween on the second end side, a semiconductor element electrically connected to the signal line on the first end side of the matching circuit by first bonding wires, a frame body, a feedthrough having a lead, and second bonding wires electrically connected to the lead and the signal line on the second end side, wherein the first bonding wires are arranged in parallel, and the second bonding wires are arranged in parallel.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a semiconductor device.

BACKGROUND ART

(2) A device that amplifies a high-frequency signal includes a semiconductor element such as a transistor, and a transmission component that transmits the high-frequency signal (PTL 1). The

transmission component has a transmission line such as a microstrip line.

## CITATION LIST

### Patent Literature

(3) PTL 1: Japanese Laid-open Patent Publication No. 63-86904

## SUMMARY OF INVENTION

### Technical Problem

(4) A characteristic impedance on one end side of the transmission component may differ from a characteristic impedance on the other end side. The change of the impedance increases the loss of the high frequency signal. Therefore, it is an object of the present disclosure to provide a semiconductor device capable of suppressing a change in characteristic impedance.

### Solution to Problem

(5) A semiconductor device according to the present disclosure includes a base; a matching circuit including: a rectangular substrate provided on the base and including an insulator; a ground layer provided on the substrate; a signal line provided on a first surface of the substrate and separated from the ground layer; wherein a width of the signal line on a first end side of the substrate is smaller than a width of the substrate and larger than a width of the signal line on a second end side of the substrate, and a distance between the ground layer and the signal line on the first end side is larger than a distance between the ground layer and the signal line on the second end side, a semiconductor element provided on the base and electrically connected to the signal line on the first end side of the matching circuit by a plurality of first bonding wires, a frame body provided on the base and surrounding the semiconductor element and the matching circuit; a feedthrough provided on the frame body and having a lead; and a plurality of second bonding wires electrically connected to the lead of the feedthrough and the signal line on the second end side of the matching circuit; wherein the plurality of first bonding wires are arranged in parallel with each other, and the plurality of second bonding wires are arranged in parallel with each other.

(6) A semiconductor device according to the present disclosure includes a base, a semiconductor element provided on a base; a feedthrough provided on a frame body and having a ground layer, and a lead provided on a signal line; wherein the semiconductor element is electrically connected to the signal line through the lead by a plurality of first bonding wires, a width of the signal line on a semiconductor element side is larger than a width of the signal line on an opposite side of the semiconductor element side, a distance between the ground layer and the signal line on the semiconductor element side is larger than a distance between the ground layer and the signal line on the opposite side of the semiconductor element side, and the lengths of the plurality of first bonding wires are the same as each other.

(7) According to the present disclosure, it is possible to suppress a change in characteristic impedance.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A is a plan view illustrating a semiconductor device according to a first embodiment.

(2) FIG. 1B is a cross-sectional view taken along a line A-A of FIG. 1A.

(3) FIG. 2A is an enlarged plan view of a matching circuit component.

(4) FIG. 2B is a cross-sectional view taken along a line B-B of FIG. 2A.

(5) FIG. 3 is a plan view illustrating the semiconductor device according to a first comparative example.

(6) FIG. 4A is a cross-sectional view illustrating the matching circuit component according to a first variation.

(7) FIG. 4B is a perspective view illustrating the matching circuit component.

(8) FIG. 4C is a cross-sectional view illustrating the matching circuit component according to a second variation.

(9) FIG. 5A is a plan view illustrating the semiconductor device according to a second embodiment.

(10) FIG. 5B is a cross-sectional view taken along a line C-C of FIG. 5A.

(11) FIG. 6 is a plan view illustrating the semiconductor device according to a second comparative example.

## DESCRIPTION OF EMBODIMENTS

### Description of Embodiments of Present Disclosure

(12) First, embodiments of the present disclosure will be listed and described.

(13) An aspect of the present disclosure is (1) a semiconductor device including a base, a matching circuit including a rectangular substrate provided on the base and including an insulator, a ground layer provided on the substrate; a signal line provided on a first surface of the substrate and separated from the ground layer; wherein a width of the signal line on a first end side of the substrate is smaller than a width of the substrate and larger than a width of the signal line on a second end side of the substrate, and a distance between the ground layer and the signal line on the first end side is larger than a distance between the ground layer and the signal line on the second end side, a semiconductor element provided on the base and electrically connected to the signal line on the first end side of the matching circuit by a plurality of first bonding wires, a frame body provided on the base and surrounding the semiconductor element and the matching circuit, a feedthrough provided on the frame body and having a lead, and a plurality of second bonding wires electrically connected to the lead of the feedthrough and the signal line on the second end side of the matching circuit, wherein the plurality of first bonding wires are arranged in parallel with each other, and the plurality of second bonding wires are arranged in parallel with each other. A characteristic impedance of the matching circuit on the one end side becomes equal to a characteristic impedance on the other end side. Therefore, the loss of the high frequency signal propagating through the matching circuit is suppressed.

(14) (2) The ground layer may be provided on a second surface of the substrate which is located on an opposite side of the first surface. The characteristic impedance of the matching circuit on the one end side becomes equal to the characteristic impedance on the other end side. Therefore, the loss of the high frequency signal propagating through the matching circuit is suppressed.

(3) The second surface of the substrate may be inclined to approach the first surface from the first end side of the substrate to the second end side of the substrate. The characteristic impedance of the matching circuit on the one end side becomes equal to the characteristic impedance on the other end side. Therefore, the loss of the high frequency signal propagating through the matching circuit is suppressed.

(4) The second surface of the substrate may have a step that approaches the first surface from the first end side of the substrate to the second end side of the substrate. The characteristic impedance of the matching circuit on the one end side becomes equal to the characteristic impedance on the other end side. Therefore, the loss of the high frequency signal propagating through the matching circuit is suppressed.

(5) The substrate may include a plurality of stacked insulator layers. The ground layer may include a first metal layer provided on a second surface of the substrate which is located on an opposite side of the first surface, and a second metal layer provided between the plurality of insulator layers. An end portion of the second metal layer on the first end side of the substrate may be located closer to the second end side of the substrate than an end portion of the first metal layer on the first end side of the substrate. The characteristic impedance of the matching circuit on the one end side becomes equal to the characteristic impedance on the other end side. Therefore, the loss of the high frequency signal propagating through the matching circuit is suppressed.

(6) The ground layer may include a plurality of stacked second metal layers. An end portion of one

of the second metal layers on the first end side of the substrate may be located closer to the second end of the substrate than an end portion of another one of the second metal layers on the first end side of the substrate. The one of the second metal layers may be closer to the signal line in a stacking direction of the plurality of insulator layers. The another one of the second metal layers may be farther from the signal line in the stacking direction. The characteristic impedance of the matching circuit on the one end side becomes equal to the characteristic impedance on the other end side. Therefore, the loss of the high frequency signal propagating through the matching circuit is suppressed.

(7) An interval between the plurality of first bonding wires may be wider than an interval between the plurality of second bonding wires.

(8) The lengths of the plurality of first bonding wires may be the same as each other, and the lengths of the plurality of second bonding wires may be the same as each other.

(9) The semiconductor device includes a semiconductor element provided on a base, a feedthrough provided on a frame body and having a ground layer, and a lead provided on a signal line, wherein the semiconductor element is electrically connected to the signal line through the lead by a plurality of first bonding wires, a width of the signal line on a semiconductor element side is larger than a width of the signal line on an opposite side of the semiconductor element side, a distance between the ground layer and the signal line on the semiconductor element side is larger than a distance between the ground layer and the signal line on the opposite side of the semiconductor element side, and the lengths of the plurality of first bonding wires are the same as each other. The characteristic impedance of the matching circuit on the one end side becomes equal to the characteristic impedance on the other end side. Therefore, the loss of the high frequency signal propagating through the matching circuit is suppressed.

#### Details of Embodiments of Present Disclosure

(15) Specific examples of a semiconductor device according to an embodiment of the present disclosure will be described below with reference to the drawings. It should be noted that the present disclosure is not limited to these examples, and is defined by Claims, and is intended to embrace all the modifications within the meaning and range of equivalency of the Claims.

#### First Embodiment

(16) (Semiconductor Device)

(17) FIG. 1A is a plan view illustrating a semiconductor device **100** according to a first embodiment. In FIG. 1A, a lid **11** described later is transparent. FIG. 1B is a cross-sectional view taken along a line A-A of FIG. 1A.

(18) The semiconductor device **100** includes a base **10**, the lid **11**, a frame body **12**, a transistor **14** (a semiconductor element), a matching circuit component **16**, and a feedthrough **18**. The base **10** is made of, for example, a stack of molybdenum/copper/molybdenum (Mo/Cu/Mo) and a metal such as a gold (Au) plating layer provided on a surface of the stack, and has a reference potential (ground potential). Sides of the base **10** extend in an X-axis direction and a Y-axis direction. A Z-axis direction is perpendicular to a top surface of the base **10**. The X-axis direction, the Y-axis direction, and the Z-axis direction are orthogonal to each other.

(19) The frame body **12** is a ring-shaped member, and is made of an insulator such as ceramic. The transistor **14** and the matching circuit component **16** are arranged on an upper surface of the base **10** at a position surrounded by the frame body **12**. Two openings are provided on both sides of the frame body **12** in the Y-axis direction, and feedthroughs **18** are attached to the two openings, respectively. Among the two feedthroughs **18**, one on the transistor **14** side is referred to as a feedthrough **18a**, and the other on the matching circuit component **16** side is referred to as a feedthrough **18b**. The feedthrough **18a**, the transistor **14**, the matching circuit component **16** and the feedthrough **18b** are arranged in this order from an upper side of FIG. 1A along the Y-axis direction.

(20) The lid **11** is attached to the upper surface of the frame body **12**. The lid **11**, the frame body **12**,

and the feedthroughs **18** hermetically seal the transistor **14** and the matching circuit component **16**. The lid **11** is made of metal such as Kovar, or ceramic. If the transistor **14** has a high moisture resistance, it does not have to be hermetically sealed. In this case, the lid **11** is made of an insulator such as ceramic or plastic.

(21) The transistor **14** is a semiconductor element including, for example, a field effect transistor (FET) using a nitride semiconductor. As illustrated in FIG. **1B**, the transistor **14** has pads **14a**, **14b** and **14c**. The pad **14a** is, for example, a source pad, and is provided on a surface on the base **10** side of the transistor **14**. The pads **14b** and **14c** are provided on a surface on the lid **11** side of the transistor **14**. The pad **14b** is, for example, a gate pad. The pad **14c** is, for example, a drain pad. A width of the transistor **14** in a longitudinal direction is preferably large for high output, and is larger than a width of a signal line **32** in the feedthrough **18**, for example.

(22) The matching circuit component **16** has a substrate **20**, a signal line **22**, and a metal layer **24**, functions as a capacitor, and matches impedances between the transistor **14** and the feedthrough **18b**. The substrate **20** is, for example, an insulating substrate made of ceramic such as aluminum oxide (Al.sub.2O.sub.3) and barium titanate (BaTiO.sub.3), and is a substrate including an insulator. The signal line **22** and the metal layer **24** are made of, for example, metals such as copper (Cu), molybdenum (Mo) and tungsten (W), and alloys containing these metals. The signal line **22** is provided on an upper surface of the substrate **20**. The metal layer **24** is a ground layer provided on a lower surface of the substrate **20**, separated from the signal line **22**, and electrically connected to the upper surface of the base **10**. The substrate **20**, the signal line **22**, and the metal layer **24** form a microstrip line that propagates a high frequency signal. An end surface on the transistor **14** side of the substrate **20** is referred to as a surface **20a**, and an end surface on the feedthrough **18b** side of the substrate **20** is referred to as a surface **20b**.

(23) FIG. **2A** is an enlarged plan view of the matching circuit component **16**. As illustrated in FIG. **2A**, a planar shape of the signal line **22** is trapezoidal. A width **W1** on the surface **20a** side of the signal line **22** is larger than a width **W2** on the surface **20b** side. The width **W1** is, for example, 4 mm, and the width **W2** is, for example, 2 mm. The width of the signal line **22** is continuously reduced from the surface **20a** to the surface **20b**. The width **W1** is, for example, 1.1 times or more, 1.5 times or more, 2 times or more, or 3 times or more, or the like of the width **W2**.

(24) FIG. **2B** is a cross-sectional view taken along a line B-B of FIG. **2A**. As illustrated in FIG. **2B**, the signal line **22** and the metal layer **24** extend in the Y-axis direction from the surface **20a** of the substrate **20** to the surface **20b**. An upper surface **20c** (first surface) of the substrate **20** and a lower surface of the metal layer **24** are parallel to the upper surface of the base **10** and extend in an XY plane. A lower surface **20d** (second surface) of the substrate **20** is inclined with respect to the surface **20c** of the substrate **20**, and approaches the surface **20c** from the surface **20a** side to the surface **20b** side. That is, the substrate **20** is continuously thinned from the surface **20a** side to the surface **20b** side. A thickness **T1** on the surface **20a** side of the substrate **20** is larger than a thickness **T2** on the surface **20b** side. The thickness **T1** is, for example, 0.5 mm, and the thickness **T2** is, for example, 0.25 mm. The thickness **T1** is, for example, 1.1 times or more, 1.5 times or more, 2 times or more, 3 times or more, or the like of the thickness **T2**.

(25) An upper surface of the metal layer **24** is inclined with respect to a lower surface of the metal layer **24**. A thickness **T3** on the surface **20a** side of the metal layer **24** is larger than a thickness **T4** on the surface **20b** side. The thickness **T3** is, for example, 0.1 mm, and the thickness **T4** is, for example, 0.35 mm. The total of the thickness of the substrate **20** and the thickness of the metal layer **24** is constant from the surface **20a** side to the surface **20b** side.

(26) As illustrated in FIG. **1B**, the feedthrough **18** is a transmission component having a substrate **30**, the signal line **32**, a metal layer **34**, and a lead **36**, and propagating a high frequency signal. The substrate **30** is, for example, an insulating substrate made of ceramic. The shape of the substrate **30** is rectangular. Upper and lower surfaces of the substrate **30** are parallel to the upper surface of the base **10** and extend in the XY plane. The metal layer **34** is a ground layer provided on the lower

surface of the substrate **30** and electrically connected to the upper surface of the base **10**. The signal line **32** and the metal layer **34** are made of, for example, metals such as Cu, Au, nickel (Ni) and titanium (Ti), and alloys containing these metals. The signal line **32** is formed on the upper surface of the substrate **30** by using, for example, sputtering and vapor deposition. The substrate **30**, the signal line **32**, and the metal layer **34** form a microstrip line. The lead **36** is made of metal, for example, is provided on an upper surface of the signal line **32**, extends in the Y-axis direction, and protrudes to the outside of the base **10**.

(27) The pad **14a** of the transistor **14** is arranged on the surface of the base **10** and is electrically connected to the base **10**. The lead **36** of the feedthrough **18a** and the pad **14b** of the transistor **14** are electrically connected by a plurality of bonding wires **40**. The pad **14c** of the transistor **14** and the signal line **22** of the matching circuit component **16** are electrically connected by a plurality of bonding wires **42**. The signal line **22** and the lead **36** of the feedthrough **18b** are electrically connected by a plurality of bonding wires **44**. The plurality of bonding wires **42** and **44** extend in the Y-axis direction. The lengths of the plurality of bonding wires **42** are equal to each other. The lengths of the plurality of bonding wires **44** are equal to each other. The bonding wires **40**, **42** and **44** are made of metals such as Au, Al and Cu.

(28) A high frequency signal having a frequency of, for example, 2 GHz is input from the lead **36** of the feedthrough **18a**, amplified by the transistor **14**, is input to the feedthrough **18b** through the matching circuit component **16**, and is output to the outside of the semiconductor device **100** through the lead **36** of the feedthrough **18b**. The feedthrough **18b** may be an input side and the feedthrough **18a** may be an output side.

(29) In order to suppress the loss of the high frequency signal, impedance matching is performed between the feedthrough **18a** and the transistor **14**, and impedance matching is performed between the transistor **14** and the feedthrough **18b**. The matching circuit component **16** is a component for the impedance matching between the transistor **14** and the feedthrough **18b**. For example, the matching circuit component **16** functions as a capacitor, and the bonding wires **44** function as an inductor.

(30) The microstrip line is formed on the matching circuit component **16** and the feedthrough **18**. The characteristic impedance of the microstrip line is determined by the width of the signal line and the thickness of the substrate (distance between the signal line and the ground layer). The wider the signal line, the lower the characteristic impedance, and the narrower the width, the higher the characteristic impedance. The larger the thickness of the substrate, the higher the characteristic impedance, and the smaller the thickness, the lower the characteristic impedance.

(31) The width of the signal line **32** of the feedthrough **18** is constant from one end to the other end in the Y-axis direction. A distance between the signal line **32** and the metal layer **34** in the feedthrough **18** in the Z-axis direction is constant. Therefore, the characteristic impedance on the transistor **14** side of the feedthrough **18** is equal to the characteristic impedance on the opposite side (a tip of the lead **36**).

(32) The width of the signal line **22** of the matching circuit component **16** decreases from the surface **20a** side to the surface **20b** side. A distance between the signal line **22** and the metal layer **24** decreases from the surface **20a** side to the surface **20b** side. Therefore, the characteristic impedance on the surface **20a** side of the matching circuit component **16** is substantially the same as the characteristic impedance on the surface **20b** side.

(33) FIG. 3 is a plan view illustrating a semiconductor device **100C** according to a first comparative example. The description of the same configuration as that of the semiconductor device **100** will be omitted. As illustrated in FIG. 3, the lower surface of the substrate **20** of the matching circuit component **16** is not inclined, and the metal layer **24** is also not inclined. The thickness of the substrate **20** is constant from the surface **20a** side to the surface **20b** side. On the other hand, the width of the signal line **22** changes between the surface **20a** side and the surface **20b** side, as in the example of FIG. 2a. The characteristic impedance of the matching circuit

component **16** changes due to the change in the width of the signal line **22**. A characteristic impedance  $Z_a$  on the surface **20a** side is lower than a characteristic impedance  $Z_b$  on the surface **20b** side, and impedance conversion is performed between the surface **20a** side and the surface **20b** side. This increases the loss of the high frequency signal propagating through the matching circuit component **16**. For example, impedance mismatch occurs between the matching circuit component **16** and the feedthrough **18b**, resulting in loss of the high frequency signal.

(34) According to the first embodiment, the substrate **20**, the signal line **22** and the metal layer **24** of the matching circuit component **16** form the microstrip line. As illustrated in FIG. 2A, the width  $W1$  on the surface **20a** side of the signal line **22** is larger than the width  $W2$  on the surface **20b** side. As illustrated in FIG. 2B, a distance  $T1$  between the signal line **22** and the metal layer **24** on the surface **20a** side (thickness of the substrate **20**) is larger than a distance  $T2$  on the surface **20b** side. Thereby, the characteristic impedance  $Z_a$  on the surface **20a** side of the microstrip line is substantially the same as the characteristic impedance  $Z_b$  on the surface **20b** side. The impedance conversion between the surface **20a** side and the surface **20b** side is suppressed, so that the loss of the high frequency signal is suppressed.

(35) The characteristic impedance  $Z_a$  and the characteristic impedance  $Z_b$  may be exactly the same or very close to each other. For example, the substrate **20** is formed of Al.sub.2O.sub.3 having a relative permittivity of 9.8, and the width  $W1$  is 4 mm, the width  $W2$  is 2 mm, the thickness  $T1$  is 0.5 mm, and the thickness  $T2$  is 0.25 mm. The characteristic impedance  $Z_a$  is  $12.11\Omega$ , and the characteristic impedance  $Z_b$  is  $12.10\Omega$ . A difference between the characteristic impedance  $Z_a$  and the characteristic impedance  $Z_b$  is, for example, 10% or less, 5% or less, 1% or less, or 0.5% or less of the characteristic impedance  $Z_a$ .

(36) As illustrated in FIG. 2A, the width of the signal line **22** is continuously reduced from the surface **20a** side to the surface **20b** side. As illustrated in FIG. 2B, the lower surface **20d** of the substrate **20** is inclined to gradually approach the upper surface **20c** from the surface **20a** side to the surface **20b**. That is, the distance between the signal line **22** and the metal layer **24** (the thickness of the substrate **20**) decreases continuously from the surface **20a** side to the surface **20b**. The width of the signal line **22** and the thickness of the substrate **20** are continuously changed, so that the characteristic impedance is substantially constant from the surface **20a** side to the surface **20b** side. The transmission line can be formed such that the characteristic impedance is not changed, and hence the loss of high frequency signals is suppressed.

(37) It is preferable to increase the width of the transistor **14** in order to increase the output. As the width increases, the width  $W1$  of the signal line **22** of the matching circuit component **16** also increases. On the other hand, the width  $W2$  of the signal line **22** is smaller than the width  $W1$  in accordance with the width of the signal line **32** of the feedthrough **18**. That is, the signal line **22** is not rectangular but trapezoidal, and the width  $W1$  and the width  $W2$  are different from each other. The signal line **22** and the transistor **14** are connected to each other by the bonding wires **42**. The signal line **22** and the feedthrough **18** are connected to each other by the bonding wires **44**.

(38) The plurality of bonding wires **42** and the plurality of bonding wires **44** extend in the Y-axis direction. The lengths of the plurality of bonding wires **42** are equal to each other, and the impedances of the plurality of bonding wires **42** are equal to each other. The lengths of the plurality of bonding wires **44** are equal to each other, and the impedances of the plurality of bonding wires **44** are equal to each other. Therefore, a phase shift between the high frequency signals is suppressed.

(39) The matching circuit component **16** may include a transmission line other than the microstrip line. For example, a strip line may be provided. The matching circuit component **16** includes the substrate and the metal layer (ground layer) on the signal line **22**. By changing the thickness of at least one of the upper and lower substrates on the surface **20a** side and the surface **20b** side, the characteristic impedances are made equal.

(40) (First Variation)



(41) FIG. 4A is a cross-sectional view illustrating a matching circuit component **16a** according to a first variation. The signal line **22** of the matching circuit component **16a** has a trapezoidal shape as in FIG. 2A. The substrate **20** of the matching circuit component **16a** is a multilayer substrate having a plurality of insulator layers **50** to **53** and metal layers **54a** to **54c**. The insulator layers **50** to **53** are stacked in order from the bottom, and adjacent insulator layers are joined to each other. The metal layer **24** is provided on the lower surface **20d** of the insulator layer **50**. The metal layer **54a** is provided between the insulator layer **50** and the insulator layer **51**. the metal layer **54b** is provided between the insulator layer **51** and the insulator layer **52**. The metal layer **54c** is provided between the insulator layer **52** and the insulator layer **53**. The signal line **22** is provided on the upper surface **20c** of the insulator layer **53**.

(42) A distance **T5** (the thickness of the substrate **20**) from the signal line **22** to the metal layer **24** is, for example, 1 mm. A distance **T6** (the total thickness of the insulator layers **51** to **53**) from the signal line **22** to the metal layer **54a** is smaller than the distance **T5**, e.g., 0.75 mm. A distance **T7** (the total thickness of the insulator layers **52** to **53**) from the signal line **22** to the metal layer **54b** is smaller than the distance **T6**, e.g., 0.5 mm. A distance **T8** (the thickness of the insulator layer **53**) from the signal line **22** to the metal layer **54c** is smaller than the distance **T7**, for example, 0.25 mm.

(43) FIG. 4B is a perspective view illustrating the matching circuit component **16**. Via holes **21** penetrating the substrate **20** in the Z-axis direction are provided. A metal layer **23** is provided on an inner wall of each via hole **21**. The metal layers **24** and **54a-54c** are electrically connected to each other by the metal layer **23** and have a reference potential (ground potential). The signal line **22** is separated from the metal layers **24** and **54a** to **54c**.

(44) As illustrated in FIG. 4A, the signal line **22**, the metal layer **24** and the metal layers **54a** to **54c** extend in the Y-axis direction. One end of each of the signal line **22**, the metal layer **24** and the metal layers **54a** to **54c** (a right end in FIG. 4A) is located on the surface **20b** of the substrate **20**. The other end of each of the signal line **22** and the metal layer **24** (the left end in FIG. 4A) is located on the surface **20a**. The other end of the metal layer **54a** is located between the surfaces **20a** and **20b**. The other end of the metal layer **54b** is located between the other end of the metal layer **54a** and the surface **20b**. The other end of the metal layer **54c** is located between the other end of the metal layer **54b** and the surface **20b**. The ends on the surface **20a** side of the metal layers **24**, **54a**, **54b**, and **54c** approach the surface **20b** in this order.

(45) According to the first variation, the characteristic impedance **Za** on the surface **20a** side of the microstrip line is substantially the same as the characteristic impedance **Zb** on the surface **20b** side, as in the first embodiment. The phase change of the high frequency signal is suppressed, and the loss is suppressed. The number of insulator layers included in the substrate **20** may be four or more, or four or less. The number of metal layers may be 3 or more, or 3 or less. Since it is not necessary to form slopes and steps in the insulators and the metal layers, the matching circuit component **16a** can be easily manufactured.

(46) (Second Variation)

(47) FIG. 4C is a cross-sectional view illustrating a matching circuit component **16b** according to a second variation. The surface **20d** of the substrate **20** of the matching circuit component **16b** has a step shape, and gradually approaches the surface **20c** from the surface **20a** side to the surface **20b** side. The metal layer **24** has a step shape corresponding to the surface **20d**.

(48) According to the second variation, the characteristic impedance **Za** on the surface **20a** side of the microstrip line is substantially the same as the characteristic impedance **Zb** on the surface **20b** side, as in the first embodiment. The phase change of the high frequency signal is suppressed, and the loss is suppressed. Any one of the substrate **20** and the metal layer **24** may have the slope and the other may have the step shape. A space between the substrate **20** and the metal layer **24** is filled with a conductive adhesive, so that the substrate **20** and the metal layer **24** are adhered to each other.

## Second Embodiment

(49) (Semiconductor Device)

(50) FIG. 5A is a plan view illustrating a semiconductor device **200** according to a second embodiment. The description of the same configuration as that of the first embodiment will be omitted. As illustrated in FIG. 5A, a planar shape of the signal line **22** of the matching circuit component **16** is, for example, a rectangle, and the width of the signal line **22** is constant from the surface **20a** side to the surface **20b** side. The planar shape of the signal line **32** of the feedthrough **18** is trapezoidal. The width on the transistor **14** side of the signal line **32** is larger than the width on the opposite side (the tip side of the lead **36**). A portion of the lead **36** that contacts the upper surface of the signal line **32** is trapezoidal like the signal line **32**.

(51) The respective bonding wires **40**, **42**, and **44** extend in the Y-axis direction. The lengths of the plurality of bonding wires **40** are equal to each other. The lengths of the plurality of bonding wires **42** are equal to each other. The lengths of the plurality of bonding wires **44** are equal to each other.

(52) FIG. 5B is a cross-sectional view taken along a line C-C of FIG. 5A. As illustrated in FIG. 5B, the upper and lower surfaces of the substrate **20** of the matching circuit component **16** are parallel to the upper surface of the base **10**, and the thickness of the substrate **20** is constant. The thickness of the metal layer **24** is also constant. A lower surface of the substrate **30** of the feedthrough **18** is inclined and approaches an upper surface thereof from the transistor **14** side to the tip end side of the lead **36**. The metal layer **34** is inclined depending on the substrate **30**.

(53) FIG. 6 is a plan view illustrating a semiconductor device **200C** according to a second comparative example. The signal line **32** and the lead **36** of the feedthrough **18** are rectangular, and the widths thereof are constant. A cross section of the semiconductor device **200C** is the same as that of the semiconductor device **100C** illustrated in FIG. 3, and a distance between the signal line **32** and the metal layer **34** of the feedthrough **18** is constant. The characteristic impedance on the transistor **14** side of the feedthrough **18** is equal to the characteristic impedance on the opposite side.

(54) As illustrated in FIG. 6, the outer ones in the X-axis direction among the bonding wires **40** and **44** extend obliquely with respect to the Y-axis direction. The ones in the center in the X-axis direction among the bonding wires **40** and **44** extend in the Y-axis direction. The outer ones among the plurality of bonding wires **40** are longer than the central one. The outer ones among the plurality of bonding wires **44** are longer than the central one. The phases of the high frequency signals are shifted in the X-axis direction due to the difference in the lengths of the bonding wires, resulting in an increase in the loss.

(55) In the second comparative example, the width of a portion of the signal line **32** increases as illustrated in FIG. 5A, so that the lengths of the plurality of bonding wires **40** and the lengths of the plurality of bonding wires **44** can be made constant. However, the change in the width of the signal line **32** causes the characteristic impedance on the transistor **14** side of the feedthrough **18** to be lower than the characteristic impedance on the opposite side.

(56) According to the second embodiment, the width on the transistor **14** side of the signal line **32** is larger than the width on the opposite side as illustrated in FIG. 5A. Therefore, the lengths of the plurality of bonding wires **40** can be made constant, and the lengths of the plurality of bonding wires **44** can be made constant. The loss of the high frequency signal can be suppressed. As illustrated in FIG. 5B, the lower surface of the substrate **30** of the feedthrough **18** is inclined. The distance between the signal line **32** and the metal layer **34** of the feedthrough **18** on the transistor **14** side is larger than the distance on the opposite side. Therefore, the characteristic impedance on the transistor **14** side of the feedthrough **18** is substantially the same as the characteristic impedance on the opposite side of the feedthrough **18**. The loss of the high frequency signal is suppressed.

(57) The substrate **30** of the feedthrough **18** may be a multilayer substrate as illustrated in FIGS. 4A and 4B. The substrate **30** and the metal layer **34** may be provided with steps as illustrated in FIG. 4C.

(58) Embodiments according to the present disclosure have been described above. However, the present disclosure is not limited to the embodiment described above, and various modifications and changes can be made to the present disclosure within the scope of the gist of the present disclosure described in the Claims.

#### REFERENCE SIGNS LIST

(59) **10** base **11** lid **12** frame body **14** transistor **14a**, **14b**, **14c** pad **16**, **16a**, **16b** matching circuit component **18**, **18a**, **18b** feedthrough **20** substrate **20a**, **20b**, **20c**, **20d** surface **21** via hole **22**, **32** signal line **23**, **24**, **34**, **54a** to **54c** metal layer **36** lead **40**, **42**, **44** bonding wire **50**, **51**, **52**, **53** insulator layer **100**, **100C**, **200**, **200C** semiconductor device

#### Claims

1. A semiconductor device comprising: a base; a matching circuit including; a rectangular substrate provided on the base and including an insulator; a ground layer provided on the substrate; a signal line provided on a first surface of the substrate and separated from the ground layer; wherein a width of the signal line on a first end side of the substrate is smaller than a width of the substrate and larger than a width of the signal line on a second end side of the substrate, and a distance between the ground layer and the signal line on the first end side is larger than a distance between the ground layer and the signal line on the second end side, a semiconductor element provided on the base and electrically connected to the signal line on the first end side of the matching circuit by a plurality of first bonding wires, a frame body provided on the base and surrounding the semiconductor element and the matching circuit; a feedthrough provided on the frame body and having a lead; and a plurality of second bonding wires electrically connected to the lead of the feedthrough and the signal line on the second end side of the matching circuit; wherein the plurality of first bonding wires are arranged in parallel with each other, and the plurality of second bonding wires are arranged in parallel with each other.
2. The semiconductor device according to claim 1, wherein the ground layer is provided on a second surface of the substrate which is located on an opposite side of the first surface.
3. The semiconductor device according to claim 2, wherein the second surface of the substrate is inclined to approach the first surface from the first end side of the substrate to the second end side of the substrate.
4. The semiconductor device according to claim 2, wherein the second surface of the substrate has a step that approaches the first surface from the first end side of the substrate to the second end side of the substrate.
5. The transmission component semiconductor device according to claim 1, wherein the substrate includes a plurality of stacked insulator layers, the ground layer includes a first metal layer provided on a second surface of the substrate which is located on an opposite side of the first surface, and a second metal layer provided between the plurality of insulator layers, and an end portion of the second metal layer on the first end side of the substrate is located closer to the second end side of the substrate than an end portion of the first metal layer on the first end side of the substrate.
6. The semiconductor device according to claim 5, wherein the ground layer includes a plurality of stacked second metal layers, an end portion of one of the second metal layers on the first end side of the substrate is located closer to the second end of the substrate than an end portion of another one of the second metal layers on the first end side of the substrate, the one of the second metal layers being closer to the signal line in a stacking direction of the plurality of insulator layers, the another one of the second metal layers being farther from the signal line in the stacking direction.
7. The semiconductor device according to claim 1, wherein an interval between the plurality of first bonding wires is wider than an interval between the plurality of second bonding wires.
8. The semiconductor device according to claim 1, wherein the lengths of the plurality of first

bonding wires are the same as each other, and the lengths of the plurality of second bonding wires are the same as each other.

9. A semiconductor device comprising: a semiconductor element provided on a base; a feedthrough provided on a frame body and having a ground layer, and a lead provided on a signal line; wherein the semiconductor element is electrically connected to the signal line through the lead by a plurality of first bonding wires, a width of the signal line on a semiconductor element side is larger than a width of the signal line on an opposite side of the semiconductor element side, a distance between the ground layer and the signal line on the semiconductor element side is larger than a distance between the ground layer and the signal line on the opposite side of the semiconductor element side, and the lengths of the plurality of first bonding wires are the same as each other.

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